

$$u_{\#}(\varphi, \theta) = \sum_{k \in \mathbb{Z}} c_k \varphi^{|k|} e^{ik\theta}$$

$$= \sum_{k \geq 0} c_k \varphi^k e^{ik\theta} + \sum_{k < 0} c_k \varphi^{-k} e^{ik\theta}$$

$$= \sum_{k \geq 0} c_k z^k + \sum_{h > 0} c_{-h} \frac{1}{z^h} \quad \begin{matrix} k = -h \\ \uparrow \\ \varphi^h e^{-ih\theta} \end{matrix}$$

$$z = \varphi e^{i\theta}$$

$$= \sum_{k \geq 0} c_k z^k + \sum_{k > 0} c_{-k} \frac{1}{z^k} \quad h \mapsto k$$

osservazione Le serie $\nearrow$ convergono nel disco $\mathbb{D}_R(0)$ uniformemente con tutte le derivate, purché la successione

$$c_k R^{|k|}$$

sia "TEMPERATA", ovvero

$$\left[\exists M > 0, m \in \mathbb{N} : |c_k R^{|k|}| \leq M(1 + |k|^m) \right]$$

Dimostrazione (per la serie $\sum_{k \geq 0} c_k z^k$)

$$|c_k z^k| = |c_k R^k \left(\frac{z}{R}\right)^k| \leq M(1 + k^m) \left|\frac{z}{R}\right|^k$$

La serie di potenze $(w := \frac{z}{R})$

$$\sum_{k \geq 0} M (1 + k^m) w^k \quad \text{converge nel disco } B_1(0) \\ (\text{unif. con tutte le derivate})$$

Infatti

$$\lim_{k \rightarrow +\infty} \sqrt[k]{M (1 + k^m)} = 1$$

Abbiamo dimostrato:

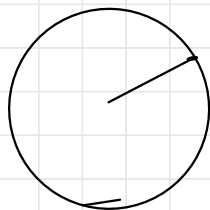
Teorema $\forall \{c_k\}$ tale che $c_k R^{|k|}$ è temperata,

la funzione

$$u_{\#}(f, \sigma) = \sum_{k \in \mathbb{Z}} c_k \rho^{|k|} e^{ik\sigma}$$

è armonica nel disco $B_R(0)$.

$$(D) \begin{cases} \Delta u = 0 & \text{in } B_R(0) \\ u(R, \theta) = g(\theta) \in L^2_{2\pi}(\mathbb{R}) \end{cases}$$



Ho, per ip., $g(\theta) = \sum_{k \in \mathbb{Z}} \hat{g}_k e^{ik\theta}$ dati!

Imponiamo

$$u_{\#}(\overset{R}{\cancel{\rho}}, \theta) = \sum_{k \in \mathbb{Z}} \underset{\substack{\uparrow \\ \text{ineognite!}}}{C_k \cancel{\rho}^{|k|}} e^{ik\theta} \quad (=) \quad \sum_{k \in \mathbb{Z}} \underset{\substack{\uparrow \\ \text{dati!}}}{\hat{g}_k} e^{ik\theta}$$

$$\Leftrightarrow C_k R^{|k|} = \hat{g}_k \quad \forall k \in \mathbb{Z}.$$

$$\boxed{C_k = \frac{\hat{g}_k}{R^{|k|}}}$$

Check: C_k temperata, ovvero $|C_k R^{|k|}| \leq M(1+|k|)^n$

vero: $g \in L^2_{2\pi}(\mathbb{R}) \Rightarrow \hat{g}_k \in \ell^2$ $\parallel \hat{g}_k \parallel$

$$\uparrow \sum_k |\hat{g}_k|^2 < +\infty$$

$$\downarrow 0 \quad |\hat{g}_k| \leq M$$

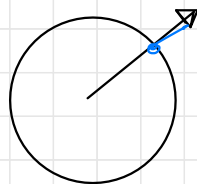
Concludere

$$u_{\#}(p, \theta) = \sum_{k \in \mathbb{Z}} \left(\frac{\hat{g}_k}{R^{|k|}} \right) \rho^{|k|} e^{ik\theta} = \sum_{k \in \mathbb{Z}} \hat{g}_k \left(\frac{\rho}{R} \right)^{|k|} e^{ik\theta}$$

$\bar{u}$ soluzione (unica) del pb. di Dirichlet

per l'eq. di Laplace sul disco.

$$(N) \quad \begin{cases} \Delta u = 0 & \text{in } B_R(0) \\ \frac{\partial u}{\partial \rho}(R, \theta) = g(\theta) \in L^2_{2\pi}(\mathbb{R}) \end{cases}$$



$$\sum_{k \in \mathbb{Z}} \hat{g}_k e^{ik\theta} \quad \text{dati!}$$

$$u_{\#}(\rho, \theta) = \sum_{k \in \mathbb{Z}} c_k \rho^{|k|} e^{ik\theta}$$

$$\frac{\partial u_{\#}}{\partial \rho}(\rho, \theta) = \sum_{k \in \mathbb{Z}} c_k |k| \rho^{|k|-1} e^{ik\theta}$$

Imponiamo:

$$\sum_{k \in \mathbb{Z}} c_k |k| R^{|k|-1} e^{ik\theta} = \sum_{k \in \mathbb{Z}} \hat{g}_k e^{ik\theta}$$

$$c_k |k| R^{|k|-1} = \hat{g}_k \quad k \in \mathbb{Z}$$

$$\nearrow k \neq 0 \Rightarrow c_k = \frac{\hat{g}_k}{|k| R^{|k|-1}} \quad \forall k \in \mathbb{Z} \setminus \{0\}$$

$$\searrow k=0 \quad \triangle!$$

$$\hat{g}_0 = 0$$

c_0 arbitrary

$$\hat{g}_0 \neq 0$$

~~no~~ solution

Conclusion:

$\rightarrow \hat{g}_0 \neq 0 \Rightarrow \nexists \text{ solution}$

$\searrow \hat{g}_0 = 0 \Rightarrow \exists \infty \text{ solutions}$

$$\begin{aligned} u \# (p, \theta) &= c_0 + \sum_{k \in \mathbb{Z} \setminus \{0\}} \frac{\hat{g}_k}{|k| R^{|k|-1}} \rho^{|k|} e^{ik\theta} \\ &= c_0 + \sum_{k \in \mathbb{Z} \setminus \{0\}} \hat{g}_k \left(\frac{\rho^{|k|}}{|k| R^{|k|-1}} \right) e^{ik\theta} \end{aligned}$$

Q. 11.

$$\hat{g}_0 = \int_0^{2\pi} g(\theta) \underbrace{e^{i0\theta}}_1 d\theta = \int_0^{2\pi} g(\theta) d\theta$$

Infatti:

$$\begin{aligned} 0 &= \int_{B_R(0)} \Delta u = \int_{B_R(0)} \operatorname{div}(\nabla u) \stackrel{\substack{\text{F.} \\ \text{G. G.}}}{=} \int_{\partial B_R(0)} \nabla u \cdot \nu \\ &= \int_{\partial B_R(0)} \frac{\partial u}{\partial \nu} = \int_{\partial B_R(0)} g \end{aligned}$$

Trasformate di Fourier in $L^1(\mathbb{R}^n)$

Def. Sia $u = u(x) \in L^1(\mathbb{R})$.

La TRASFORMATA DI FOURIER di u è

la funzione

$$\hat{u}(\xi) = \int_{\mathbb{R}} u(x) e^{-i\xi x} dx \quad \forall \xi \in \mathbb{R}$$

Osservazioni

- L'integrale è in " dx "
- $\left| \int_{\mathbb{R}} u(x) e^{-i\xi x} dx \right| \leq \int_{\mathbb{R}} |u(x) e^{-i\xi x}| dx$
 $= \int_{\mathbb{R}} |u(x)| \underbrace{|e^{-i\xi x}|}_{=1} dx \quad (e^{i\theta} = \cos\theta + i\sin\theta)$
 $= \int_{\mathbb{R}} |u(x)| dx < +\infty$
- $e^{-i\xi x} = \cos(-\xi x) + i \sin(-\xi x)$
 $= \cos(\xi x) - i \sin(\xi x)$

$$\hat{u}(\xi) = \int_{\mathbb{R}} u(x) (\cos(\xi x) - i \sin(\xi x)) dx$$

$$= \underbrace{\int_{\mathbb{R}} u(x) e^{i\xi x} dx}_{\operatorname{Re}(\hat{u}(\xi))} + i \underbrace{\left[- \int_{\mathbb{R}} u(x) \operatorname{Im}(\xi x) dx \right]}_{\operatorname{Im}(\hat{u}(\xi))}$$

• Generalizzazioni: si può partire invece che da $u \in L^1(\mathbb{R})$

$$\rightarrow u \in L^1(\mathbb{R}; \mathbb{C}) \quad \int_{\mathbb{R}} |u|_1 < +\infty$$

$$\rightarrow u \in L^1(\mathbb{R}^n)$$

$$\hat{u}(\xi) = \int_{\mathbb{R}^n} u(x) e^{-i\xi \cdot x} dx$$

pr. scalare in $\mathbb{R}^n$
 $(x_1 \xi_1 + \dots + x_n \xi_n)$.

$$\rightarrow u \in L^1(\mathbb{R}^n; \mathbb{C}) \quad \int_{\mathbb{R}^n} |u|_1 < +\infty$$

• Analogia coi coeff. di Fourier

$$\hat{u}_k = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} u(x) e^{-i\xi_k x} dx \quad u \in L^2_T(\mathbb{R})$$

$$= \frac{1}{T} \frac{1}{2\pi} \int_{-\pi}^{\pi} u(x) e^{-i k x} dx \quad u \in L^2_{2\pi}(\mathbb{R})$$

$k \in \mathbb{Z}$

$$\hat{u}(\xi) = \int_{\mathbb{R}} u(x) e^{-i\xi x} dx \quad \xi \in \mathbb{R}$$

- La TRASFORMATTA DI FOURIER è anche l'operatore che manda u in $\hat{u}$.

$$\mathcal{F}: u = u(x) \longrightarrow \hat{u} = \hat{u}(\xi)$$

$\hat{} \quad \quad \quad \textcircled{?}$

$L^1(\mathbb{R}^n)$

→ è un operatore lineare su $L^1(\mathbb{R}^n)$

$$\mathcal{F}(\alpha u + \beta v) = \alpha \mathcal{F}(u) + \beta \mathcal{F}(v)$$

$\parallel \qquad \qquad \qquad \parallel$

$$\int_{\mathbb{R}} (\alpha u(x) + \beta v(x)) e^{-i\xi x} dx \qquad \alpha \hat{u}(\xi) + \beta \hat{v}(\xi)$$

$\parallel \qquad \qquad \qquad \parallel$

$$\int_{\mathbb{R}} \alpha u(x) e^{-i\xi x} + \beta v(x) e^{-i\xi x} \stackrel{\text{linearità}}{=} \alpha \int_{\mathbb{R}} u(x) e^{-i\xi x} dx + \beta \int_{\mathbb{R}} v(x) e^{-i\xi x} dx$$

Trasformate notevoli (Parte I).

$$\uparrow \quad u(x) = \chi_{(a,b)}(x) = \begin{cases} 1 & x \in (a,b) \\ 0 & x \notin (a,b) \end{cases}$$

$$\begin{aligned} \hat{u}(\xi) &= \int_{\mathbb{R}} \chi_{(a,b)}(x) e^{-i\xi x} dx \\ &= \int_a^b e^{-i\xi x} dx = \int_a^b (\cos(\xi x) - i \sin(\xi x)) dx \\ &= \left. \frac{\sin(\xi x)}{\xi} + i \frac{\cos(\xi x)}{\xi} \right|_a^b \end{aligned}$$

$$\hat{u}(\xi) = \left[\frac{\sin(\xi b) - \sin(\xi a)}{\xi} \right] + i \left[\frac{\cos(\xi b) - \cos(\xi a)}{\xi} \right]$$

In particolare $a = -b$, $u(x) = \chi_{(-b,b)}(x)$

$\hat{u}$ è puramente reale

$$\hat{u}(\xi) = \frac{2 \sin(\xi b)}{\xi}$$

$$b = 1 \quad u = \chi_{(-1,1)}$$

$$\hat{u}(\xi) = \frac{2 \sin \xi}{\xi}$$

Commento: In generale
 $\hat{u} \notin L^1(\mathbb{R}^n)$.

$$2) \quad u(x) = e^{-|x|} = \begin{cases} e^{-x} & x \in \mathbb{R}^+ \\ e^x & x \in \mathbb{R}^- \end{cases}$$



$$\hat{u}(\xi) = \int_{\mathbb{R}} e^{-|x|} e^{-i\xi x} dx = \int_{-\infty}^0 e^{x(1-i\xi)} dx + \int_0^{+\infty} e^{-x(1+i\xi)} dx$$

$$= \frac{e^{x(1-i\xi)}}{(1-i\xi)} \Big|_{-\infty}^0 + \frac{e^{-x(1+i\xi)}}{-(1+i\xi)} \Big|_0^{+\infty}$$

$$= \frac{1}{1-i\xi} + \frac{1}{1+i\xi} = \frac{\cancel{1-i\xi} + \cancel{1+i\xi}}{1+\xi^2} = \frac{2}{1+\xi^2}$$

$$\hat{u}(\xi) = \frac{2}{1+\xi^2}$$

(in q.s. caso $\hat{u} \in L^1(\mathbb{R})$).

$$3) \quad u(x) = \frac{1}{1+x^2}$$

$$\hat{u}(\xi) = \int_{\mathbb{R}} \frac{1}{1+x^2} e^{-i\xi x} dx \quad -\xi = \omega$$

$$\leadsto \int_{\mathbb{R}} \frac{1}{1+x^2} e^{i\omega x} dx \quad \text{integrali di tipo 3}$$

$$\hat{u}(\xi) = \pi e^{-|\xi|}$$

$$\left[\begin{array}{c} \text{commento} \quad \hat{u} \in L^1(\mathbb{R}) \\ e^{-|x|} \xrightarrow{\mathcal{F}} \frac{2}{1+\xi^2} \xrightarrow{\mathcal{F}} (2\pi) e^{-|x|} \end{array} \right]$$

$$?? \rightarrow \hat{\hat{u}} = (2\pi)u \leftarrow ??$$

$$\left\{ \begin{array}{l} \hat{u}_k \text{ coeff di Fourier di } u \leadsto u(x) = \sum_k \hat{u}_k e^{ikx} \\ \hat{u} \text{ transf di Fourier di } u \leadsto u \quad (??) \end{array} \right.$$

PROPRIETÀ DI $\mathcal{F}$ $\begin{cases} \rightarrow \text{analisi funzionale} \\ \rightarrow \text{algebraiche} \\ \rightarrow \text{differenziali} \\ \rightarrow \text{antitransformata} \end{cases}$

Teorema di Riemann-Lebesgue

Sia $u \in L^1(\mathbb{R})$, Allora:

1) $\hat{u} \in L^\infty(\mathbb{R})$

[$\mathcal{F}: L^1(\mathbb{R}) \rightarrow L^\infty(\mathbb{R})$ op. lineare]
continua di norma 1

2) $\hat{u}$ continua

3) $\hat{u}$ "infinitesimo all'infinito" $\lim_{\xi \rightarrow \pm\infty} \hat{u}(\xi) \rightarrow 0$

Dim.

1) $|\hat{u}(\xi)| = \left| \int_{\mathbb{R}} u(x) e^{-i\xi x} dx \right| \leq \int_{\mathbb{R}} |u(x)| dx = \|u\|_1$

$$\Rightarrow \|\hat{u}\|_\infty \leq \|u\|_1$$

$$\left(\Rightarrow \|\mathcal{F}\|_{\mathcal{L}(L^1, L^\infty)} \leq 1 \right) \text{ (in realtà vale =)}$$
$$\inf \{ M: \|\mathcal{F}(u)\|_\infty \leq M \|u\|_1 \}$$

2) $\hat{u}$ continua in $\xi \in \mathbb{R}$ (ξ qualsiasi)

$$\xi_m \rightarrow \xi \stackrel{?}{\Rightarrow} \hat{u}(\xi_m) \rightarrow \hat{u}(\xi)$$

$$\hat{u}(\xi_m) = \int_{\mathbb{R}} \underbrace{u(x) e^{-i\xi_m x}}_{f_m(x)} dx \stackrel{?}{\rightarrow} \hat{u}(\xi) = \int_{\mathbb{R}} \underbrace{u(x) e^{-i\xi x}}_{f(x)} dx$$

$$u(x) e^{-i\xi x} \longrightarrow u(x) e^{-i\xi x} \quad \text{q.o. } x \in \mathbb{R}$$

$$u(x) [\cos(\xi x) - i \sin(\xi x)] \quad u(x) [\cos(\xi x) - i \sin(\xi x)]$$

$$|f_n(x)| = |u(x)| \quad u \in L^1(\mathbb{R})$$

$$\text{per conv. dominata, } \int_{\mathbb{R}} f_n(x) dx \longrightarrow \int_{\mathbb{R}} f(x) dx$$

$$3) \bullet u = \chi_{(a,b)} \Rightarrow \text{VERO } \left(\lim_{\xi \rightarrow \pm\infty} \hat{u}(\xi) = 0 \right)$$

$$\bullet u = \sum_i c_i \chi_{(a_i, b_i)} \Rightarrow \text{VERO } \left(\hat{u} = \sum_i c_i \widehat{\chi_{(a_i, b_i)}} \right)$$

$$\lim_{\xi \rightarrow \pm\infty} \hat{u} = \sum_i c_i \underbrace{\lim_{\xi \rightarrow \pm\infty} \widehat{\chi_{(a_i, b_i)}}}_{=0} = 0$$

$$\bullet \text{ data } u \in L^1(\mathbb{R}), \exists \{u_n\} \text{ "a scalino" : } u_n \xrightarrow{0} u \text{ in } L^1(\mathbb{R})$$

(dalle teoreme di approssimazione con $C_0^\infty(\mathbb{R})$).

$$u_n \xrightarrow{L^1} u \Rightarrow \hat{u}_n \xrightarrow{L^\infty} \hat{u}$$

$$\text{Per ogni } n, \text{ se che } \lim_{\xi \rightarrow \pm\infty} \hat{u}_n(\xi) = 0$$

$$\Rightarrow \text{anche } \lim_{\xi \rightarrow \pm\infty} \hat{u}(\xi) = 0$$

$\uparrow$
es.

□

Proprietăți algebrice $\hat{f} \in L^1(\mathbb{R})$

• $v(x) = u(x-a) \quad a \in \mathbb{R}$

$$\Rightarrow \hat{v}(\xi) = e^{-i\xi a} \hat{u}(\xi)$$

$$\left(\int_{\mathbb{R}} \underbrace{u(x-a)}_{\text{"y"}} e^{-i\xi x} dx \right)$$

• $v(x) = e^{iax} u(x) \quad a \in \mathbb{R}$

$$\Rightarrow \hat{v}(\xi) = \hat{u}(\xi - a)$$

$$\left(\int_{\mathbb{R}} e^{iax} u(x) e^{-i\xi x} dx \right)$$

• $v(x) = u\left(\frac{x}{a}\right) \quad a \in \mathbb{R} \setminus \{0\}$

$$\Rightarrow \hat{v}(\xi) = |a| \hat{u}(a\xi)$$

$$\left(\int_{\mathbb{R}} \underbrace{u\left(\left(\frac{x}{a}\right)\right)}_{\text{"y"}} e^{-i\xi x} dx \right)$$

Un particular : ~~cu~~ $a = -1$

$$\begin{cases} u \text{ par } (u(x) = u(-x)) \Rightarrow \hat{u} \text{ par } (\text{e real}) \\ u \text{ impar } (u(x) = -u(-x)) \Rightarrow \hat{u} \text{ impar } (\text{e puramente imaginare}) \end{cases}$$

Esempi Calcolare $\int \dots$

$$\bullet \frac{\cos x}{1+x^2} \quad \frac{1}{1+x^2} e^{ix} = \frac{\cos x}{1+x^2} + i \frac{\sin x}{1+x^2}$$

$$\bullet \frac{1}{1+(x+3)^2} \quad \frac{1}{1+x^2}$$

$$\bullet \frac{1}{4+x^2} \quad \frac{1}{4\left(1+\left(\frac{x}{2}\right)^2\right)}$$